Counting

onsdag den 13. november 2024 12.30

Counting

when k+1 or more objects are placed into k boxes there exits at least 1 box with at least 1 or more objects

General:

when N or more objects are into k boxes there exits at least 1 box containing $\lceil \frac{N}{k} \rceil$ objects

division rule

$$n = \frac{|A|}{|B_i|}$$

$a \cdot b \quad a + b$

$\wedge \quad \vee$

Permutationer/kombinatorik

P Rækkefølge vigtigt

C Rækkefølge ikke vigtigt

$$P(n,r) = \frac{n \cdot (n-1) \cdot (n-2) \cdots 2 \cdot 1}{(n-r) \cdots 2 \cdot 1} = \frac{n!}{(n-r)!}$$

$$= n(n-1) \cdots (n-r+1)$$

$$C(n,r) = \frac{P(n,r)}{r!} = \frac{\frac{n!}{(n-r)!}}{r!} = \frac{n!}{(n-r)! \cdot r!}$$

$0 \leq r \leq n$

n = Antal objecter

r = udtag

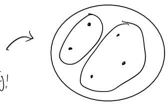
$$\frac{8!}{(8-4)! \cdot 4!} = \frac{40.320}{576} = 70$$

$$\frac{6!}{(6-3)! \cdot 3!} = \frac{720}{36} = 20$$

$$\frac{8!}{(8-2)! \cdot 2!} \quad \frac{40320}{1440} = 28$$

Identity

$$\binom{n}{r} = \binom{n}{n-r} \stackrel{def}{=} \frac{n!}{n(n-r))! \cdot (n \cdot r)!} = \frac{n!}{r! (n-r)!}$$



$$\sum_{k=0}^{n} \binom{n}{k} = 2^n$$

Pascals Identity

$$\binom{n+1}{k} = \binom{n}{k-1} + \binom{n}{k}$$

Binomial Coefficient

$\binom{n}{r}$ (n chose r)

$$(x+y)^2 = x^2 + 2xy + y^2 = \binom{2}{0}x^2 + \binom{2}{1}xy + \binom{2}{2}y^2$$

$$(x+y)^3 = (x+y) \cdot (x^2 + 2xy + y^2)$$

$$= 1x^3 + 3x^2y + 3xy^2 + 1y^3$$

$$= \binom{3}{0}x^3 + \binom{3}{1}x^2y + \binom{3}{2}xy^2 + \binom{3}{3}y^3$$

$$(x+y)^n = \sum_{j=0}^{n} \binom{n}{j} \cdot x^{n-j} y^j$$

Repetition & Indistinguishability:

Permutation & Repetition

n^r

Combination & Repetition

$$\binom{n+(r-1)}{r}$$

$* \; | ** | * | **$

n = 4

r = 6

COOKIE

$$\frac{P(6,6)}{P(2,2)} = \frac{6!}{2!}$$

INDISTINGUISHABILITY

$$\frac{P(20,20)}{P(6,6)\, P(2,2)^3} \quad \frac{20!}{6! \cdot (2!)^3}$$

$$\frac{n!}{n_1! \cdot n_2! \cdots n_k!}$$

$$\boxed{\begin{array}{l} \text{distribute n distinguishable objects to m indistinguishable boxes} \\ \sum_{j=1}^{k} \frac{1}{j!} \sum_{i=0}^{j-1} (-1)^i \binom{j}{i} (j-i)^n \end{array}}$$